

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Zenith

Team Name: Byte Me

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian university students particularly those at the University of Malaya are surrounded by a genuine wealth of opportunities: scholarships, internships, competitions, grants, certifications, and career programmes. Despite this abundance, most students still miss out because the information is scattered across websites, emails, social media, university channels, and word-of-mouth. The real problem is not a shortage of opportunities. It is a shortage of clarity, prioritization, and execution. Even when students find opportunities, a long list tells them nothing about which one fits their course, year, CGPA, goals, readiness, and available time or what to do about it right now.
- **Target Domain:** Domain 2 AI for Economic Empowerment & Decision Intelligence. Zenith enables university students to make smarter, faster, and better-informed decisions about which opportunities are worth pursuing, with Z.AI's GLM as the central reasoning engine powering all scoring, explanation, and advisory functions.
- **Proposed Solution Summary:** Zenith is an AI-powered opportunity platform that helps students discover, evaluate, prioritise, and act on scholarships, internships, grants, competitions, certifications, and career programmes in one place. Instead of showing students a long list of announcements, Zenith explains which options fit them best, why they matter, what they are missing, and what action they should take next. Powered by the scoring engine which combines fit scoring, ROI scoring, urgency scoring, readiness scoring, and strategic value, Zenith turns decision overload into a clear weekly action plan.

2. Background & Business Objective

- **Background of the Problem:** Students search across websites, emails, social media, university channels, and word-of-mouth just to find scholarships, internships, grants, competitions, and programmes. When they do find something, a long list of opportunities does not tell them which one fits their course, year, skills, goals, CGPA range, readiness, and available time. High-value opportunities are then missed because students do not manage deadlines, documents, preparation tasks, and application progress in one connected system. This fragmented landscape creates four compounding pain points which are scattered information across channels that is fragmented and time-consuming to check; no personal priority signal because students do not know which opportunity fits them best; readiness gaps where missing documents and requirements silently block strong applications; and weak execution because deadlines, tasks, and follow-ups are not managed in one place. The result is measurable economic loss. Every semester, students miss scholarships worth tens of thousands of ringgit, internships with RM 3,000–7,000 monthly stipends, and competitions with significant prize pools, not because the opportunities did not exist, but because no intelligent system helped them find the right ones, understand their eligibility, and act before the deadline.
- **Importance of Solving This Issue:** Solving this is not just a convenience improvement. It is an economic empowerment problem. A student who captures one additional relevant scholarship or internship per semester that they would otherwise have missed gains hundreds to thousands of ringgit in direct financial value. At the University of Malaya scale, the collective economic impact across the student body is substantial. Zenith's Insights page makes this concrete by tracking total accessible opportunity value per student, and value at risk of expiring.
- **Strategic Fit / Impact:** Zenith aligns precisely with Domain 2's requirements: AI for economic empowerment, where GLM is central to interpreting data, reasoning with context, generating recommendations, and explaining decisions clearly with measurable economic impact. Without the scoring engine, Zenith collapses into a static directory. With it, Zenith becomes a genuine decision advisor where every recommendation comes with a score, a reason, a risk, and a next step. The platform is designed to start with the University of Malaya as the pilot community, then scale nationally across Malaysian universities, and eventually grow into a regional student opportunity ecosystem that connects students, universities, and opportunity providers in one platform.

3. Product Purpose

3.1. Main Goal of the System: Zenith's goal is to close the gap between a student's awareness of opportunities and their ability to act on the right ones at the right time. It does this not by showing more listings, but by reasoning over them, scoring fit, urgency, ROI, effort, readiness, and strategic value then telling the student exactly what to do this week, what to prepare for next, and what to skip.

3.2. Intended Users (Target Audience):

- University of Malaya students
- Students actively seeking scholarships, internships, competitions, startup grants, and certifications
- Students with limited time who need clear, prioritized guidance on where to invest their effort
- Universities seeking scalable career advising and student readiness analytics
- Opportunity providers - scholarship bodies, companies, grant organisations that seeking verified, targeted distribution

4. System Functionalities

4.1. Description: Zenith is a full-stack web application. The frontend is a React Router application with pages covering the dashboard, opportunities, pipeline, planner, AI advisor, profile, settings, and search. The backend is a FastAPI application with routes supporting every major feature area: profile, dashboard, opportunities, pipeline, readiness, planner, advisor, documents, and settings. The AI and data layer is built around four core services: CatalogService (opportunity data management), ScoringService (the scoring engine), GLMService (the Z.AI GLM / ILMU integration layer), and ScreenService with JsonStateStore (state and screen management). The opportunity dataset is a curated, seeded JSON store for the prototype, with PostgreSQL planned for production.

4.2. Key Functionalities:

Feature / Module	Description
Dashboard	Personalised command centre showing top matches, expiring value at risk, readiness score, weekly insights and the AI copilot entry point.
Opportunity Discovery	Live-ranked list of scholarships, internships, grants, competitions, certifications, and programmes, filtered and sorted by Best Match, Highest ROI, Most Urgent, or Low Effort. Each card shows fit score, ROI score, effort level, deadline, estimated RM value, and a GLM-generated 'Why recommended' explanation.
Decision Pipeline	Kanban-style tracker with four stages: Apply Now, Prepare Soon, Track Later, Skip for Now. Displays total estimated value, number of high-fit opportunities, active applications, and at-risk deadline alerts.
Applications Tracker	Tracks submitted applications with submission date, deadline, current stage (Application Submitted / Under Review / Accepted), and application progress bar.
Calendar	Deadline and interview calendar synced with the opportunity pipeline, with upcoming events panel showing application deadlines, interview slots, and planner tasks tagged by type.
Documents Manager	Upload and organize application documents: CV/Resume, Academic (transcript), Certifications, Personal, and Others. Shows document readiness status (Ready / Needs Update) and how many applications each document is linked to.
Network	Connect with mentors, alumni, and peers. Shows suggested connections based on shared career interests and opportunity pipeline, with direct messaging.
AI Advisor	Conversational AI interface where students ask strategic questions.

Feature / Module	Description
Resource Hub	Library of guides, templates, videos, and webinars. For example, CV writing guide, scholarship application masterclass, interview preparation checklist, portfolio guides, personal statement templates.
Readiness Tracker (Opportunity Readiness)	Scores the student's asset readiness: CV/Resume, Transcript, Portfolio, LinkedIn, Essay/Personal Statement, Referee, Certifications, Profile Completeness. Shows overall readiness score, impact on top opportunities, and AI-generated improvement checklist.
Weekly Priority Planner	AI-generated weekly action plan with tasks sorted into Apply Now, Prepare Soon, Track Later, and Skip for Now columns. Each task shows estimated time, ROI classification, and link to the related opportunity. Includes an Optimize Plan feature and AI Planning Tips panel.
Insights & Economic Impact	Dashboard showing total accessible opportunity value, value at risk of expiring, high-priority opportunity count, time saved vs manual search, match-to-goals percentage, missed value tracker, opportunity categories by RM value, deadline urgency breakdown, readiness bottlenecks, and pipeline growth over time.
Student Profile	Academic profile management: full name, email, phone, university, course, faculty, year, goal, CGPA range, areas of interest, time availability, readiness level. Includes an AI Profile Summary panel showing match quality, top matched opportunities, and a strength breakdown.
Settings	Account management, notification preferences, AI preferences, language and region, privacy and security, subscription management (Pro Scholar Plan), data controls, and help and support.

4.3. AI Model & Prompt Design

4.3.1 Model Selection

Zenith uses Z.AI's GLM (GLM / ILMU integration) as its core AI engine. GLM was selected because it delivers the natural language reasoning quality required to interpret unstructured opportunity descriptions, assess profile-to-opportunity fit across multiple dimensions simultaneously, and produce clear, plain-language explanations that students can trust and act on.

4.3.2 Prompting Strategy

Zenith uses a multi-step, structured prompting approach. Step 1 is a profile parsing prompt that converts the student's onboarding inputs into a structured context object. Step 2 is an opportunity analysis prompt that passes this context alongside each opportunity record and asks GLM to score the opportunity across

all six dimensions and generate a recommended action, reasoning summary, risk flag, and next step. Step 3 is a priority synthesis prompt that organizes all scored opportunities into the four pipeline stages with a weekly action plan. For the AI Advisor chat, the strategy is context-injected conversational prompting: the student's full profile, current pipeline state, and readiness status are passed as system context before each user query, enabling the GLM to give strategically grounded answers rather than generic advice.

4.3.3 Context & Input Handling

Student profiles are structured into approximately 800 tokens of context. Opportunities are analyzed in batches of up to 20 per GLM call. Opportunity descriptions exceeding 400 tokens are summarized before being passed. The AI Advisor maintains a rolling 10-turn context window. GLM integration includes an AI cache layer to avoid re-scoring unchanged opportunities on repeat sessions, reducing both cost and latency. Structured JSON output is requested from GLM for all scoring calls to enable reliable parsing and rendering.

4.3.4 Fallback & Failure Behaviour

If the live GLM API is unavailable or returns an unusable response, Zenith falls back to grounded explanations generated from the scoring data alone. The fallback layer uses pre-computed score values and rule-based logic to produce a reduced but still coherent recommendation. This is surfaced transparently to the user. The system does not silently fail, every recommendation card is either AI-generated with full GLM reasoning or clearly marked as a computed estimate. The GLM integration also includes observability and test coverage, which means the team can monitor AI response quality and catch degradation early.

5. User Stories & Use Cases

- **User Stories:**

- As a Year 2 Computer Science student, I want to see a ranked list of internships and scholarships matched to my CGPA, course, and goals, so I can stop spending an hour every week manually checking different websites and still feel unsure.
- As a student with limited time, I want to know which 3 opportunities deserve my attention this week, with a reason for each, so I can invest my effort where it counts most.
- As a student who just found an opportunity close to its deadline, I want to instantly see my fit score, eligibility gaps, and a recommended action, so I can decide in seconds whether to rush an application.
- As a student with gaps in my application documents, I want to know exactly which missing assets are blocking my highest-value opportunities, so I can prioritise what to prepare first.
- As a student managing multiple applications, I want a pipeline view that tracks the stage of each application and shows expiring deadlines, so I never miss a critical date.
- As a student unsure whether to prioritise a high-value scholarship or a fast-to-win competition, I want the AI Advisor to compare them and explain the trade-off, so I can make a confident decision.

- **Use Cases (Main Interactions):**

Use Case	Description
Profile Onboarding	Student completes a short setup form in under 2 minutes: university, course, faculty, year, CGPA range, areas of interest, primary goal, time availability, readiness level, and available assets (CV, transcript, portfolio, LinkedIn). This data becomes the primary input for all scoring and GLM reasoning.
Opportunity Discovery & Ranking	Student opens the Opportunities page. The engine scores all opportunities in the curated database against the student's profile and returns a ranked list. Each card displays fit score, ROI score, effort level, estimated RM value, deadline, eligibility status, and a GLM-generated 'Why recommended' explanation with two or three reasoning bullets. Student can sort by Best Match, Highest ROI, Most Urgent, or Low Effort, and filter by category.

Use Case	Description
Opportunity Deep Dive	Student clicks any opportunity card. The full detail view shows: title, source, category, deadline, estimated RM value, fit score, urgency score, ROI score, effort level, eligibility confidence, recommended action (Apply Now / Prepare Soon / Track Later / Skip), GLM reasoning summary, risk or missing requirement, next step, and a direct link to the official opportunity page.
Pipeline Management	Student views the Pipeline page. Opportunities are sorted across four Kanban columns: Apply Now (high-fit, high-urgency), Prepare Soon (strong match but needs readiness work), Track Later (relevant but lower urgency), and Skip for Now (low-fit or low-value). Each column shows cards with value, fit score, effort level, and deadline. Student can move cards between columns and mark opportunities as applied.
Applications Tracking	Student tracks submitted applications on the Applications page. Each entry shows submission date, deadline, current stage (Application Submitted, Application Review, Accepted), and a visual progress bar. Tabs allow filtering by All, In Progress, Submitted, Under Review, Accepted, and Rejected.
AI Advisor Chat	Student types a question or taps a quick prompt such as 'Help me plan my next 7 days', 'Which opportunities should I prioritise?', 'Show me high ROI opportunities', or 'What scholarships am I eligible for?'. The GLM responds using the student's full profile, pipeline state, and readiness status as context. The AI advisor also surfaces active goals and tracks progress against them in the sidebar.
Readiness Tracker	Student opens the Readiness page. An overall readiness score (e.g. 67%) is shown alongside scores for each asset module: CV/Resume (90%, High Impact), Transcript (82%, High Impact), Portfolio (48%, High Impact), Essay/Personal Statement, Referee, Certifications, and Profile Completeness. GLM generates an AI checklist showing the top readiness improvements ranked by their impact on blocking opportunities, including which specific opportunities each improvement would unlock.
Weekly Priority Planner	Student opens the Planner. GLM has generated a current-week action plan showing the total estimated time (e.g. 6.6 hrs), focus score, and tasks across four columns: Apply Now (high ROI), Prepare Soon (medium ROI), Track Later (lower urgency), and Skip for Now (low ROI). Each task links to its related opportunity. The student can tap Optimise Plan to trigger a GLM re-run, add custom tasks, or view the AI Planning Tip panel which explains why the plan is structured the way it is.
Insights & Economic Impact	Student views the Insights page showing: total accessible opportunity value (e.g. RM 727,000), value at risk expiring soon (e.g. RM 228,500), high-priority opportunities identified, time saved in manual searching (e.g. 79.2 hours), match-to-goals percentage, missed value tracker, opportunity categories by RM value breakdown (Scholarship, Internship, Programme, Grant, Competition, Certification), deadline urgency distribution, readiness bottlenecks bar chart, and pipeline growth over time. The student can export a report.

Use Case	Description
Document Management	Student uploads CV, transcript, cover letters, portfolio, recommendation letters, and other documents. The system shows readiness status, file size, creation date, and the number of applications each document is linked to. Missing or outdated documents are flagged.
Network & Connections	Student views current connections (mentors, alumni, peers) and receive GLM-suggested connections based on their career goals and active opportunities. The student can message connections and receive guidance on specific applications.
Resource Hub	Student browses guides, templates, videos, and webinars. A search bar helps locate specific resources. Upcoming webinars are surfaced in the sidebar. A prompt directs students to the AI Advisor for personalized resource guidance.

6. Features Included (Scope Definition)

- Student profile onboarding covering academic details, goals, constraints, and asset readiness
- Curated, seeded JSON opportunity database of verified Malaysian scholarships, internships, competitions, grants, certifications, and career programmes
- Scoring engine with Fit Score, Urgency Score, ROI Score, Effort Level, Readiness Score, and Eligibility Confidence for every opportunity
- GLM / ILMU explanation layer generating reasoning bullets, trade-offs, next step, and confidence flags for every recommendation
- AI Explanation fallback to deterministic grounded explanations if the live GLM API fails
- Personalized dashboard with pipeline value summary, expiring value alert, weekly insights, and AI copilot entry
- Four-column decision pipeline: Apply Now, Prepare Soon, Track Later, Skip for Now
- Opportunity discovery page with multi-dimension sorting (Best Match, Highest ROI, Most Urgent, Low Effort) and category filters
- Application tracker with stage tracking (Submitted, Under Review, Accepted) and progress bars
- Calendar view with deadline and interview dates synced to the pipeline
- Document manager for CV, transcript, certifications, personal documents, and others with readiness status
- Network page with mentor, alumni, and peer connections and suggested connections

- AI Advisor chat with context-injected conversational prompting, quick-prompt buttons, and active goals panel
- Resource Hub with guides, templates, videos, and webinars plus an upcoming webinars panel
- Opportunity Readiness Tracker scoring all major asset categories with an AI-generated improvement checklist
- Weekly Priority Planner with AI-generated action tasks, time estimates, ROI classification, and Optimize Plan feature
- Insights & Economic Impact page with missed value tracker, pipeline growth charts, readiness bottlenecks, and export report
- Student profile management with AI Profile Summary, match quality score, and strength breakdown
- Settings with notification preferences, AI preferences, language and region, privacy controls, and subscription management
- Pro Scholar Plan subscription tier with unlimited AI Advisor queries, priority matching, advanced analytics, and custom templates
- AI cache layer to avoid re-scoring unchanged opportunities on repeat sessions

7. Features Not Included (Scope Control)

- Real-time web scraping or automated opportunity ingestion from external sources, the prototype uses a curated seeded JSON dataset
- Direct integration with external university student portals, government scholarship systems, or company HR platforms
- Native iOS or Android mobile application, the current version is a responsive web application
- Multi-user simultaneous collaboration or team application features
- Automated application form submission on behalf of students
- AI-generated CV or personal statement drafting, Zenith advises on readiness but does not write documents for students
- Video interview preparation, mock interview simulation, or video recording features
- Integration with LinkedIn, Glassdoor, or other third-party job boards

- Production-grade PostgreSQL database integration, the prototype uses the seeded JSON data layer and JsonStateStore
- Two-factor authentication or enterprise-grade security features

8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated token usage per average user session is approximately 3,000–5,000 tokens, covering the profile context, opportunity batch scoring (up to 20 items), and a typical AI Advisor conversation of 3–5 turns. Two key design decisions manage inference costs at scale: first, the AI cache layer prevents re-scoring of opportunities that have not changed since the last session; second, the AI Advisor chat uses a rolling 10-turn context window rather than the full conversation history, significantly reducing tokens per message. Structured JSON output is requested from GLM for all scoring calls to minimize verbose free-form responses.
- **Technical Constraints:**
 - The prototype uses a seeded JSON dataset of curated Malaysian opportunities, real-time ingestion and automated scraping are not implemented
 - The GLM / ILMU API must be available and responsive, extended downtime triggers the deterministic fallback explanation layer
 - The React Router frontend is optimized for desktop browsers, mobile responsiveness is supported but the primary design target for the prototype is desktop
 - Document analysis is limited to metadata and readiness status tracking, automated CV parsing or content analysis is not included in scope
 - Full frontend-to-backend integration with live authentication is identified as the primary remaining engineering step
- **Performance Constraints:**
 - Opportunity scoring for a batch of 20 items targets a GLM response time of under 8 seconds, larger batches use progressive loading
 - AI Advisor chat targets a response time of under 5 seconds per turn
 - The AI cache layer ensures repeat sessions for unchanged profiles are significantly faster as GLM calls are skipped for cached scores

- **User Input Constraints:**

- Recommendation quality depends on the accuracy of the student's onboarding profile, incomplete or inaccurate profile data degrades scoring quality
- The platform relies on the student to upload their own documents, it cannot verify the authenticity or quality of uploaded assets
- Opportunity data is curated by the Byte Me team and requires manual refreshing, opportunities may become outdated between refresh cycles

9. Risks & Questions Throughout Development

- **GLM API Reliability for Live Demo:** The live demo depends on GLM / ILMU API stability. Zenith mitigates this with a full fallback layer that serves deterministic, grounded explanations when the live API fails. The AI cache layer also means previously scored results can be served without a live API call, making the demo resilient even under partial API degradation.
- **Opportunity Data Staleness:** The curated JSON dataset can become outdated as deadlines pass and new opportunities emerge. For the prototype, the team manually verifies each entry before seeding. For production, a scheduled refresh mechanism and eventual institutional data feeds are planned. The calendar and deadline alert features also surface expiring items prominently to reduce the impact of any lag.
- **GLM Hallucination in Scoring or Explanation:** GLM may occasionally generate an inaccurate fit score justification or misinterpret an eligibility requirement. This is mitigated through structured JSON output requests with explicit scoring criteria in the prompt, a confidence score returned with every AI response, transparent display of uncertain fields, and a visible disclaimer on every recommendation card that students should verify eligibility with the official source.
- **Profile Accuracy Dependency:** If a student provides inaccurate profile information, recommendations will be miscalibrated. The AI Profile Summary page addresses this by showing students how their profile data is driving their match quality score, creating an incentive to keep the profile accurate. The strength breakdown (Completeness, Assets, Relevance, Engagement) also helps students identify profile weaknesses.